

Smart VAT

Smart VAT is a tool to convert FBX and Alembic animations to Vertex Animation Textures.

Supported Platforms

Built and tested in Unity Version 2022.3.10 but can probably work on other versions.

Supported Render Pipelines

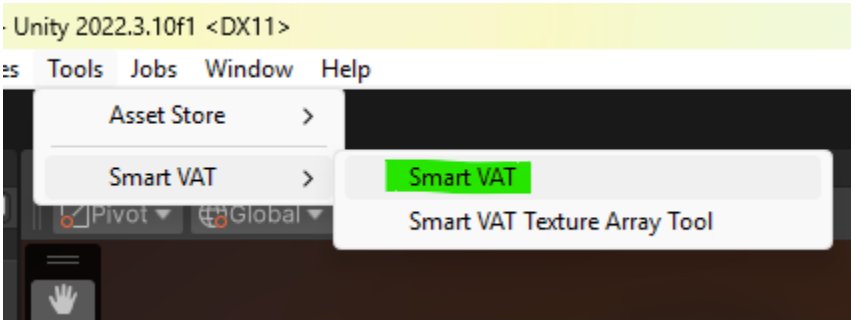
URP

BiRP - **IMPORTANT: Only works if Color Space is Linear for EXR textures to be interpreted properly**

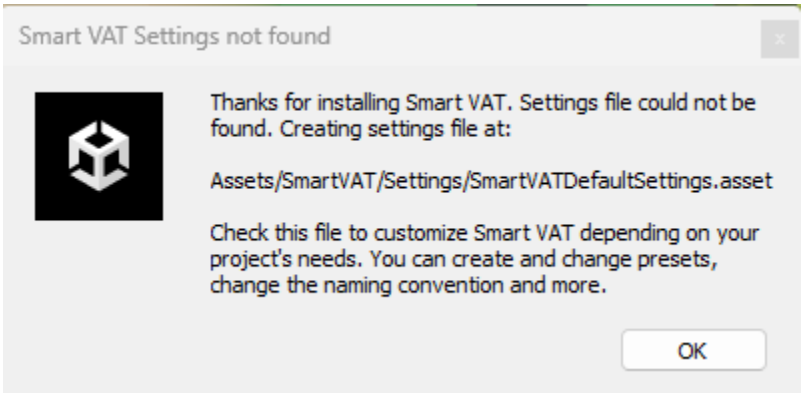
HDRP

After installing for the first time

Open the Smart VAT Window from the Tools menu at the top.



When launching Smart VAT for the first time, the tool will create a Settings File in the Assets Folder. It only happens if you don't have that file already, preventing your personal settings from getting crushed.

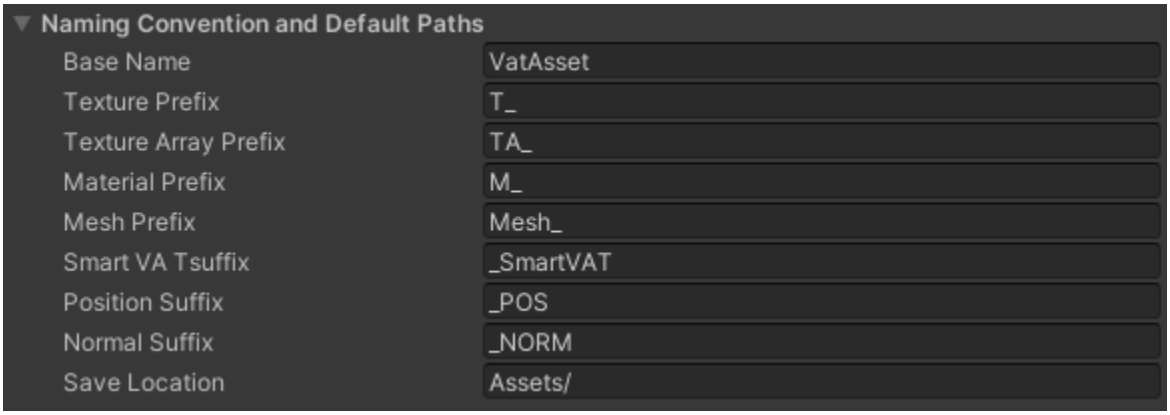


Configuring Smart VAT

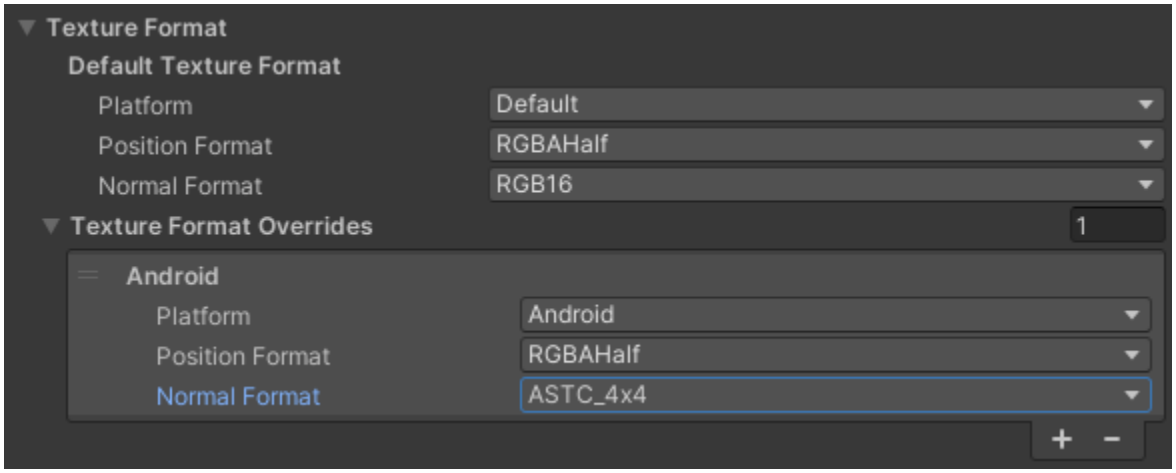
Selecting the Settings file at Assets/Plugins/SmartVAT/Settings/SmartVATDefaultSettings.asset will reveal all of the customizable options in the inspector.

The default options should work right out of the box.

The options of the first section are all related to Naming Conventions and Default Paths



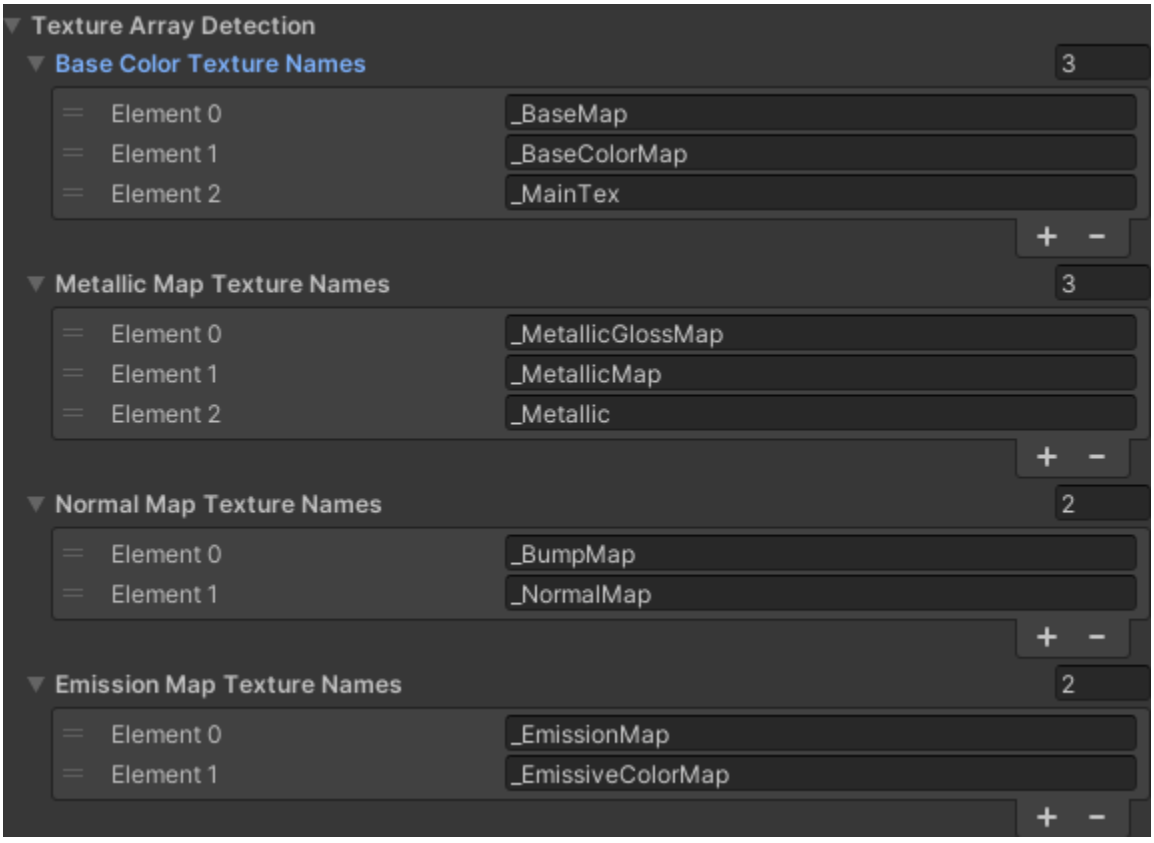
The second section is all about texture file formats. If you install modules for other platforms, like Android and Apple, you'll be able to set the Texture Format Overrides here. Note: It won't affect files that have already been created, only future ones.

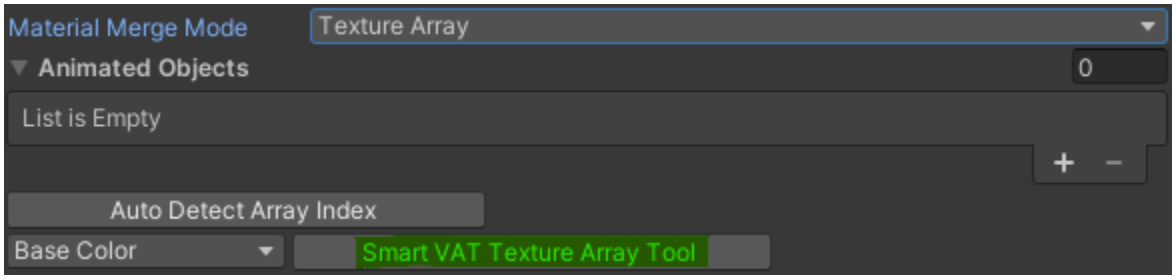


The debug material is the material of the Filter Mesh Preview. It helps visualize the parts of the mesh that won't generate texture data.

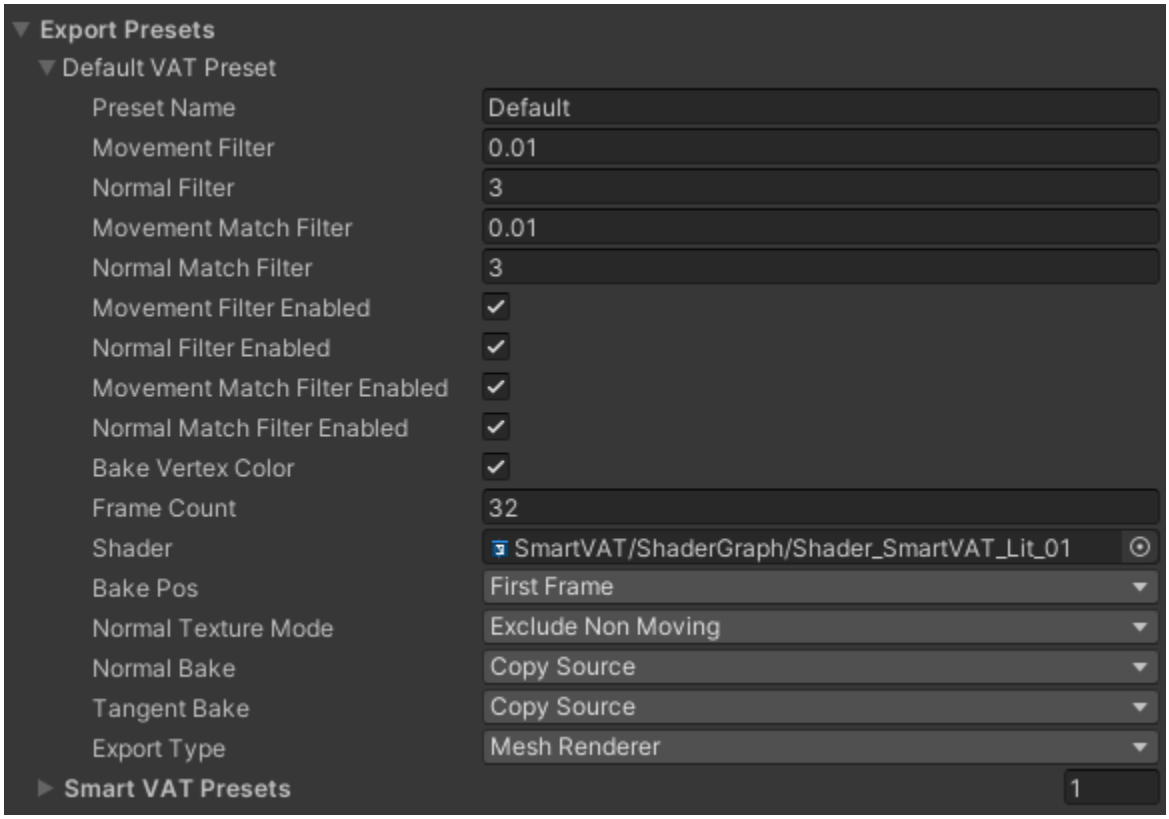


Texture Array Detection is to be used with the Texture Array Tool. When merging multiple objects, you might be dealing with multiple materials. Those settings determine what Material Property name the tool should look for in order to attempt to create the array.



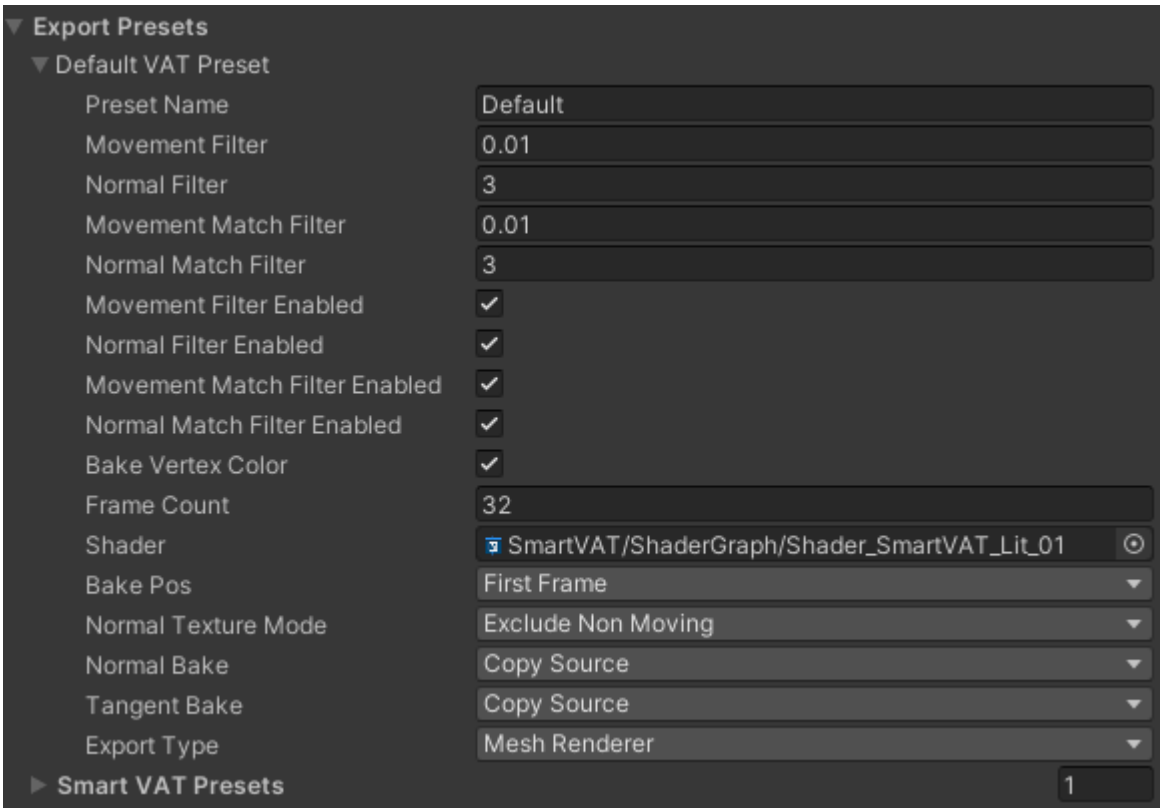


Export Presets are to make combinations of settings easy to reuse.



Note: After loading a preset in the smart VAT Window, you can show the Advanced Settings and tweak the numbers as you want. It won't affect the preset.

Settings



Movement Filter

Any vertex that moves by a distance lower than that amount will get skipped and won't generate any movement texture data. Ideal for removing slight shakiness.

Normal Filter

Any normal that rotates by a value lower than this angle will get skipped and won't generate any normal texture data.

Movement Match Filter

Vertices whose movement are always within that tolerance will share the same movement texture data.

Normal Match Filter

Normals whose rotations are always within that tolerance will share the same normal texture data.

Bake Vertex Color

Choose whether you want to copy over the vertex colors from the original mesh or not. Setting this off simply returns white vertex colors for the entire mesh.

Shader

If you created your own compatible VAT shader to use along with this tool, you can set it here as a default. The VAT material will automatically be created with this shader. Please note that the code depends on specific material property names. You'll have to keep the same names otherwise the textures won't get automatically assigned.

Bake Pos

Position in which the mesh gets baked.

> **Original Pose:** Default mesh will be in the original position without animation, like a Tpose.
>
> **First Frame:** Default Mesh will take the shape of the first frame of the animation.

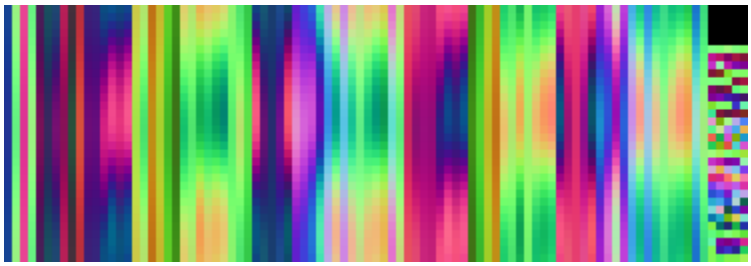
Normal Texture Mode

> **Exclude Non Moving:** Normal Data is only created for moving vertices.
>
> **Include All:** One single extra pixel is added (as opposed to a whole column of animation data) for each non-moving normal in the mesh. This setting exists because when spawning meshes from a Particle System, all meshes spawned from it lose any kind of info about their local orientation, making the shader deform the models in the wrong axes. A hack to re-create that orientation info is to bake the right vector of the mesh into the vertex normals, hence the need to bake even non-moving normals to the texture.

Export Type

> **Mesh Renderer** automatically creates an object with a Mesh Filter and Mesh Renderer components with the right files associated with each.
>
> **Particle System** creates a Particle System with all of the right Vertex Streams in order for the orientation info to be restored. The Material also gets created with the property "_ParticleVertexStream" automatically on as well as "_Animation_Mode" set to 2, which means that it is controlled through the Particle's Custom1 Module curve X.

Example of a Normal Texture set to Include All. Each smooth column of the texture to the left is animated data of a normal. The very end to the right where the pixels are very different from each other are individual non-animated normals. Doing it this way saves a lot of texture space.



Using Smart VAT

Smart VAT requires the user to set up a few things in a scene in order to use.

A. Add animated meshes to the Animated Objects list

1- Drag the Mesh Renderer/Skinned Mesh Renderer game objects from the scene hierarchy into Animated Objects

2- Next to Playable Director, click “Create New from Animated Objects”, this will prompt you to save the timeline asset somewhere and automatically link the Game Object in the Timeline.

3- Add the animation(s) that you want to bake to the track next to your object and scrub the animation to

ensure that it is properly playing.

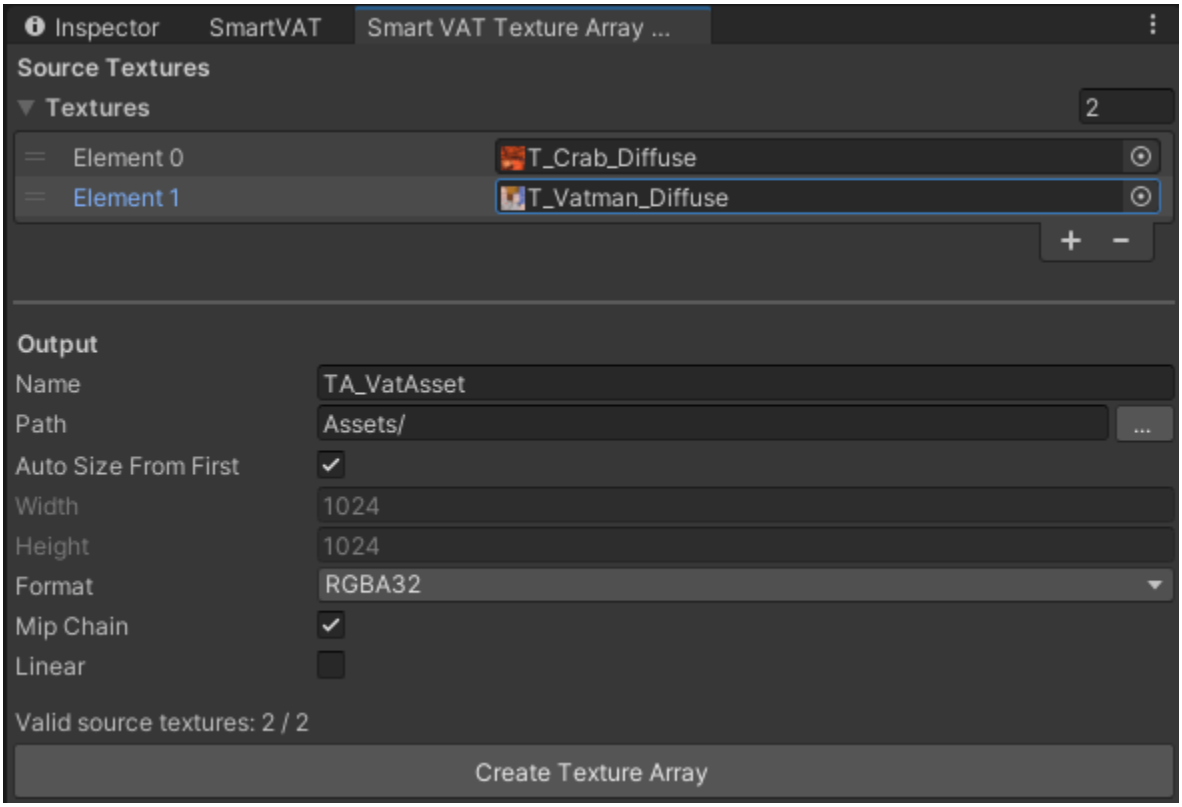
OPTIONAL. Create Texture Array

When merging multiple meshes, for example, a character that is made of different objects for their body and their hair.

1- Choose the Texture Map lookup, for example: Base Color.

2- Click Auto Detect Array Index: This makes the tool search for all of the materials assigned to the meshes and look for the texture. If 2 materials share the same texture, they will keep the same array index.

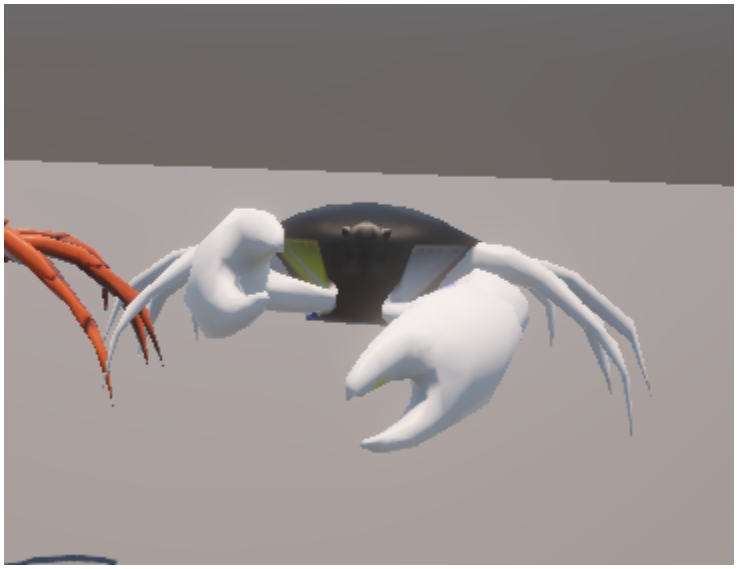
3- Click Smart VAT Texture Array Tool. This will take you to another window where you can pack multiple textures into an array to be used later after the VAT Export. You'll have to choose an appropriate shader that supports texture arrays.



B. Preview optimization

1- Choose a Preset and tweak advanced settings if needed.

2- Click Filter Mesh Preview. This will create a temporary mesh that helps visualize the optimization provided by Smart VAT by coloring areas that won't generate texture data.



Black: Doesn’t move and normals don’t rotate.

Yellow: Moves but normals don’t rotate.

Blue: Doesn’t move but normals rotate.

C. Exporting

1- Set the desired Asset Base Name and Export path.

2- Click Export VAT.

Important Note: The result might appear instantly or take several minutes based on the total number of vertices and frames. The tool doesn't impose any hard limits. More vertices and frames mean bigger textures.

Technical Information

Some of the info below has already been stated above, but here is a more complete list of technicalities regarding the tool.

Output textures resolutions are like this:

Position Texture

X resolution = Number of moving vertices

Y resolution = Number of frames

Normal Texture

X resolution = Number of moving normals + non moving normals(also packed vertically in available space) when normal mode is set to Include All.

Y resolution = Number of frames

UV0 is automatically copied to the end result mesh.

UV1.x:

VAT Position Texcoords X + No Movement Filter Mask.

UV1.y:

VAT Normal Texcoords X + No Rotate Normal Filter Mask

UV2.x:

VAT Normal Texcoords Y (used for Include All normals only) + Texture Array Index

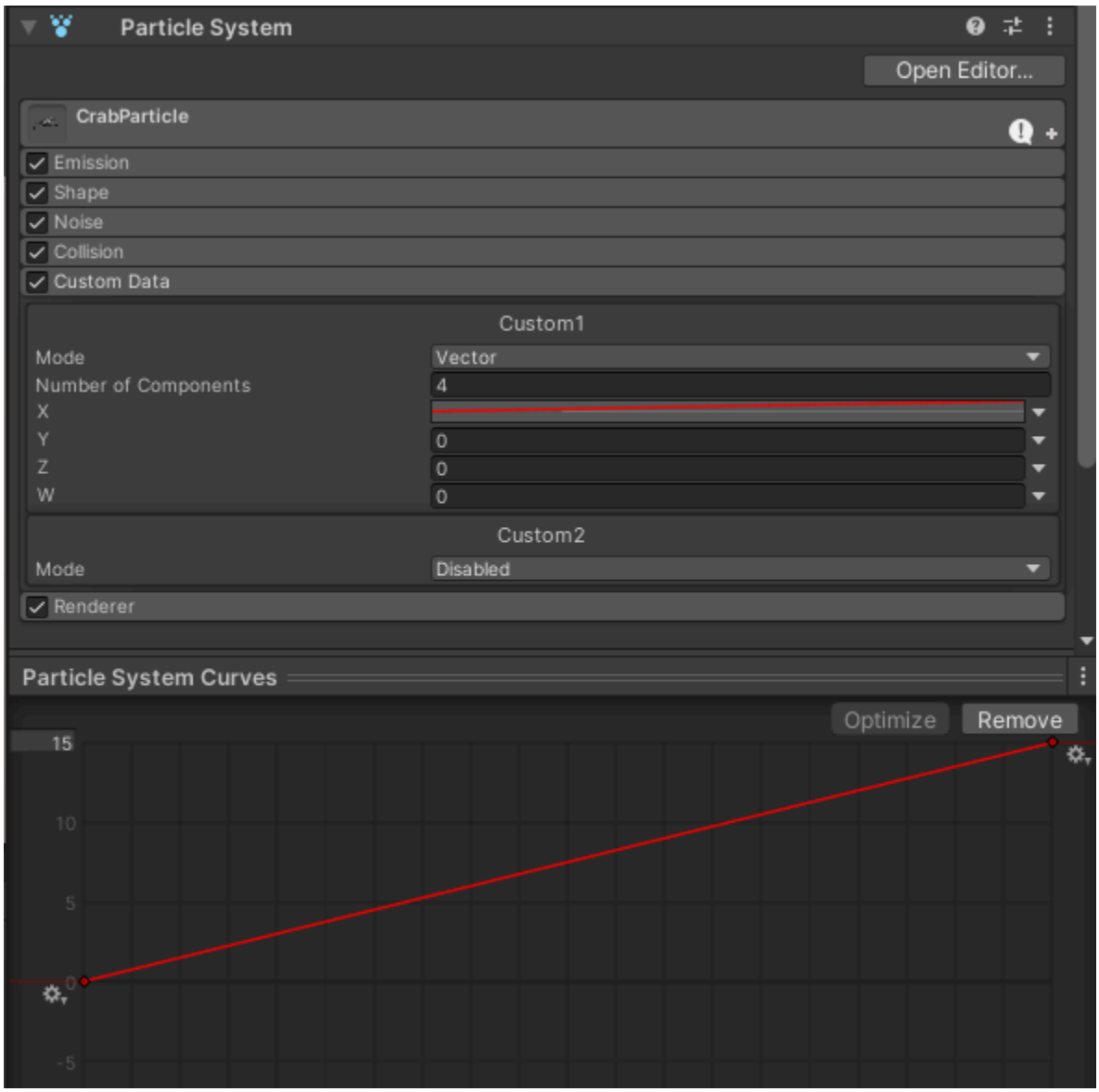
For animated positions and normals, they don't need a baked y coordinate because this corresponds to the timing of the animation which is controlled by a different parameter.

Animation Mode 0: Automatic, influenced by Auto Animation Speed.

Animation Mode 1: Manual, must be animated by the Manual Time property. Can also be controlled from an animation.

Animation Mode 2: Baked into UV3.x which corresponds to Particle Custom1.x.

The curve below loops the animation 15 times over the life of the particle.



BiRP issue

Your project's color space has to be Linear in order to be able to read the EXR textures.

Smart VAT's code looks for and modifies the following material properties. If you create your own version of the shader, make sure to follow the same naming convention or else you'll have to modify the code.

_VATPosition

_VATNormal

_Animation_Mode

_ParticleVertexStream

Note about particles:

When using particles, UVs are not automatically carried over unless Custom Vertex Streams are enabled. On top of this, they follow a slightly different convention where the 3rd and 4th channels of the UV vector are used instead of being ignored. Because of this, the shader needs the material property "_ParticleVertexStream" in order to be able to properly play the animation.

Particles need the following Vertex Streams in order to be compatible with the Smart VAT default shader.



Note: The tool currently adds Custom1.xyzw and Custom2.xyzw. Those are kept there for possible future features.

Meshes emitted from particles don't preserve any information about their orientation in the world. This causes shader deformations to deform meshes in the wrong axes. There are 2 things however that particle systems orient properly based on the mesh's rotation and they are Vertex Normals and Vertex Tangents. In order to communicate with the shader the right axes in which to deform the meshes, Vertex Normals have to be baked to X+ (right) and Vertex Tangents have to be baked to Z+ (forward). The up vector is calculated in the shader by doing a cross product between them.

Until Unity can provide a more efficient way to preserve a particle mesh's orientation, we have no choice but to rely on this hack. Fortunately, the tool comes with a way to bake the otherwise destroyed normals to the output texture.

